When and How Often is Weighted Majority Vote Optimal Under Log Loss?

Steven An [1] Sanjoy Dasgupta [1]

[1]Department of Computer Science and Engineering
University of California, San Diego

Abstract

Suppose one has labeling functions (LFs, e.g. "document contains 'birdie' ⇒ 'sports' is its topic") that predict on unlabeled datapoints. How should one combine their predictions to get a single labeling when LFs have different accuracies and are sometimes contradictory? One strategy is to do a weighted majority vote over their predictions. We study the optimality of weighting strategies under log loss, i.e. functions that take LF accuracies, predictions, etc., and return LF weights.

Formal Model

- n unlabeled datapoints $X = \{x_1, x_2, \ldots, x_n\} \subset \mathcal{X}$
- k labels denoted $\mathcal{Y} = \{1, 2, \ldots, k\}$
- p labeling functions (LFs):

$$h^{(1)}, h^{(2)}, \ldots, h^{(p)} \colon X \to \mathcal{Y}$$

Goal: For each datapoint x_i and labels $1 \leq \ell \leq k$, infer the underlying conditional label probability

$$\eta_{i\ell} = \Pr(y_i = \ell \mid x_i).$$

- We'll estimate this using (a function of) a weighted majority vote over the LF predictions.

A First Pass Approach: Majority Vote (MV)

Each LF receives weight 1. This does well when the majority of LFs predict the correct label, and is often good in practice.

Some Failure Modes:

- Large class imbalance and no highly accurate LFs
  - Better to predict the majority class and ignore LFs altogether
- Large number of spammers (LFs with accuracy $\approx 1/k$)
  - If there are only a few high accuracy LFs, their votes get drowned out by the spammers' votes (as each spammer gets weight 1)

An Attempted Fix: One-Coin Dawid-Skene

For this strategy, LF $h^{(j)}$ gets weight θ_j proportional to its accuracy and class ℓ gets weight ω_ℓ proportional to its frequency.

- Let $b_j^* = Pr(h^{(j)}(x) = y), w_\ell^* = Pr(y = \ell)$ and

$$\theta_j = \log\left(\frac{b_j^*(k-1)}{1 - b_j^*}\right), \qquad \omega_\ell = \log w_\ell^*$$

- The OCDS estimate of $\eta_{i\ell}$ is $Pr(y = \ell \mid h^{(j)}(x_i), \forall j \in [p])$

$$\propto \exp\left(\omega_\ell + \sum_{j=1}^{p} \theta_j \mathbf{1}(h^{(j)}(x_i) = \ell)\right)$$

How does this address MV's modes of failure?

- Large class imbalance and no highly accurate LFs
  - For any minority class ℓ s.t. $w_\ell^* \ll 1$, the class weight $\omega_\ell \ll 0$
- Large number of spammers (LFs with accuracy $\approx 1/k$)
  - $\theta_j \approx 0$ whenever $b_j^* \approx \frac{1}{k}$ so spammers get little weight

Problem: These θ_j, ω_ℓ are almost never optimal w.r.t. log loss!

Simple and Complex Weighting Strategies

Weighting strategies are functions convert information such as LF accuracies, class frequencies, etc., into weights.

Simple Weighting Strategies

These strategies *only* use LF accuracies and class frequencies:

$$\theta \colon [0,1]^p \times \Delta_k \to \mathbb{R}^p, \quad \omega \colon [0,1]^p \times \Delta_k \to \mathbb{R}^k$$

- Majority Vote: $\theta = \mathbf{1}_p, \omega = \mathbf{0}_k$
- One-Coin Dawid-Skene: $\theta = \log\left(\frac{b(k-1)}{1-b}\right), \omega = \log(w)$

Complex Weighting Strategies

These weighting strategies use LF accuracies, class frequencies, *and* the LF predictions. This difference is crucial!

Our Contributions

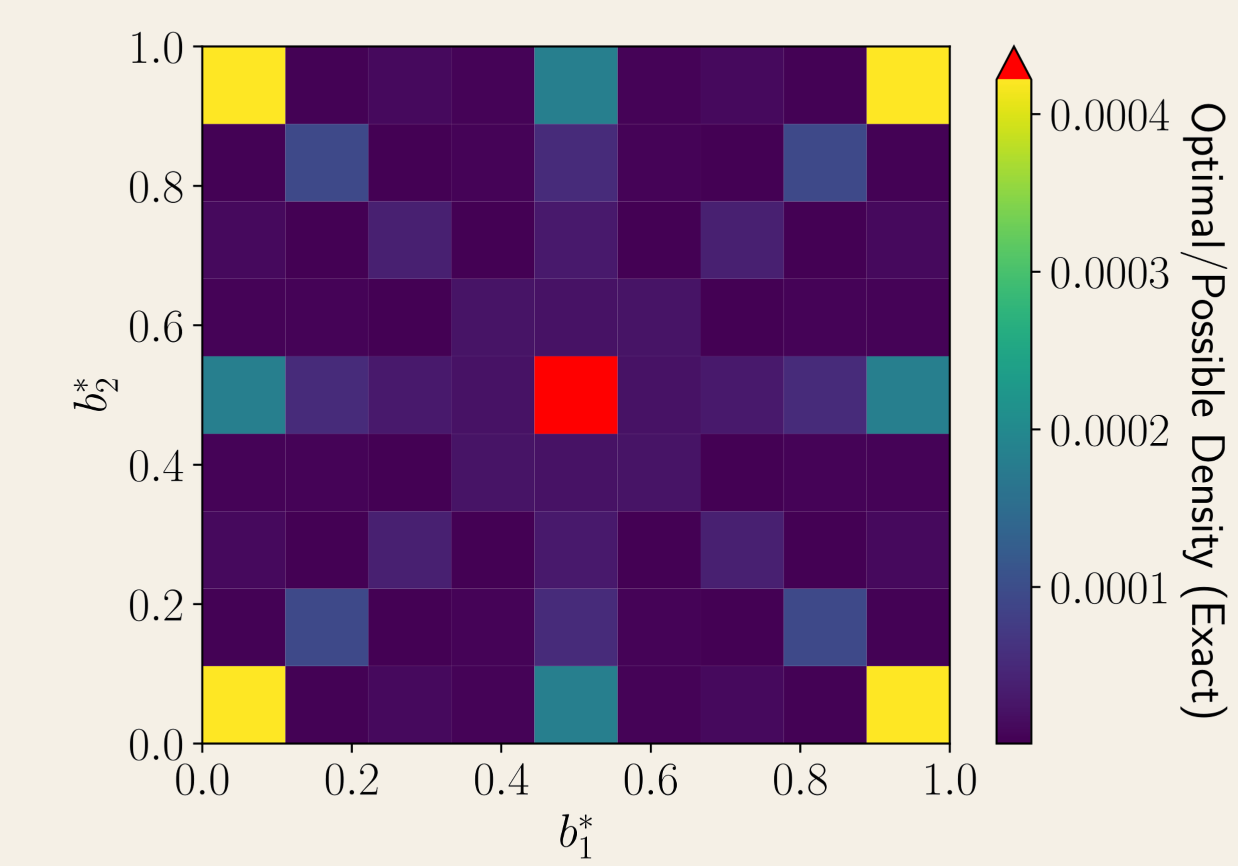
Optimality of Simple Weighting Strategies

Fix LF accuracies b^*, class frequencies w^*, and say we estimate η using the OCDS functional form. Then, simple weighting strategies are almost never optimal w.r.t. log loss.

Why? Too many bad cases. E.g. simple weighting strategies give duplicated LFs weight proportional to # of duplicates.

How do we avoid this? Use complex weighting strategies, e.g. Balsubramani-Freund, if one requires optimality.

Experimental Result For $p = k = 2, w^* = \mathbf{1}_2/2$, we compute the proportion of problems where OCDS is optimal.



Exact Analysis of Error from Weighted Majority Vote

Consider two weighting strategies (θ, ω) and (θ', ω').

Weighting Strategy Comparison An exact expression for the extra error incurred by (θ, ω) over (θ', ω') is given. We also give a test to determine whether this quantity is > 0.

Covariate Shift Analysis Suppose (θ, ω) is better than (θ', ω'). We exactly characterize how much covariate shift can be tolerated before (θ', ω') becomes better than (θ, ω).